

Comparative Study of LI-FI and WI-FI: A Review

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ARTICLE INFO

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DOI:

10.24081/nijesr.2017.1.0006

Keywords:

VLC; LI-FI; WI-FI; Security; and line of sight.

ABSTRACT

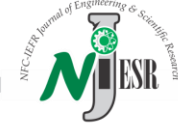
The battle between light and radio waves are not new. Alexander Grahams Bell also used radio signal for propagation of sound. Although these radio waves signals don't equal to the light waves with respect to speed. With the advancement in technology this can be possible. Now signals are not only travel with the speed of light but also communicate with the help of light. Everyone knows about WI-FI but now the era of LI-FI has been start. With the help of visible light communication, now the idea of communication through light is possible in the form of LI-FI. LI-FI and WI-FI basically the two sides of the same coin therefore these two technologies can be compared. This research article compares the two technologies for delivering wireless internet access services i-e; WI-FI vs. LI-FI. Although the above two wireless technology has different medium but same services. The LI-FI recently gets an attention of world. Some communication platform predict that this new technology definitely replaces the WI-FI in very near future. This paper presents the comparison between these two technologies. Different parameters like medium, range, direction and security are used to suggest that which one is better in different environment for diverse users. The upcoming challenges of LI-FI and its applications are also discussed. At the end, critical discussion is provided to new way towards this new technology for new research.

I. INTRODUCTION

The role of technology is rapidly increased in every field of life. The technology may be in the form of wired and wireless medium. But in the modern era everyone prefers the wireless medium as compared to wired one. In 2015, the million numbers of users in this world that attach with wireless technology but according to the latest survey of wireless communication 210 million of internal users present inside the world. An according the recent survey that are conducted between wired and wireless technology, the number of users of wireless networks has increased ratio therefore wireless networks play part and parcel role in every fields of networks. The most well-known example of wireless network is WI-FI and LI-FI [1]. The chairman of mobile communication professor herald Hass [2] give concept of transmission of data using LED bulbs [3]. In 2011, he coined the term LI-FI at university of Edinburgh at TED global talk and Start Company to market it. In 2012 LUMIX is an organization that owns this technology privately in California. The first smart phone prototype of LI-FI was presented in front of consumer electronically show in the city of Las Vegas in 2014 [4]. The basic operation of WI-FI transmission is performed with the help of radio waves. The WI-FI routers play our role for

transmission of data but have interference issue when nearby access points are present. On the other hand LI-FI, because its name light fidelity so its transmission with the help of light i-e LED bulbs so LI-FI [5] have not any issue of interference similar to radio frequency waves. The major technology used in LI-FI and WI-FI is present IrDA and WLAN 802.11a/b/g/n/ac/ad and some standard complaint devices respectfully. Due to this technology being used in LI-FI the number of application are present like in airline cabins, underwater exploitations, hospitals, education sectors, power plants, energy sectors, street lamps, vehicle to vehicle communication, control traffic conditions and mean internet everywhere because light present everywhere. On the other hand, the application of WI-FI nothing but used for internet browsing with the help of WI-FI hotspots. At above large areas of LI-FI application it has maximum advantages like it works in dens region with less interference also solve the problems that are directly or indirectly related to inefficiency of radio frequency bandwidth therefore it has high data transmission rates.

The LI-FI [6] mechanism totally worked on light waves and light can't pass through walls and does not penetrate some other material so this technology gives us security and privacy. No one knows about either someone used LI-FI or not. In this case,



no interference is possible. LI-FI completely based on light in the form of visible light communication (VLC) so no radiation are present being used LI-FI [3]. But in case of WI-FI interference is present and always has some privacy issue because radio signal does not block by the walls. The speed of data transfer in LI-FI has about 1Gbps and WI-FI has 1-2Gbps can be achieved using Giga-1R and WiGig but overall WLAN-11n offer 150Mbps with frequency of 2.4-4.9 GHz and 5 GHz. Thus LI-FI have 10 thousand times frequency spectrum of the radio waves that is the main reason LI-FI has high density as compared to WI-FI. WI-FI coverage area is about 32 meters approximately and it is vary based on antenna type power transmission. The complete LI-FI system consist of LED lamp driver and photo detector, while on other hand the WI-FI must be required routers to be installed from some subscriber device stations. Another fundamental reason LI-FI over WI-FI is that about health risk. All the wireless devises have some smart chip that emits different type of harmful radiations. If thicknesses of the wireless devices under an extended period then scientists believe that it may create tumors that have special name of disease called insomnia. As the number of users is increased day-by-day, the WI-FI pollution is increased. Due to WI-FI pollution increased, it can change the sleep patterns and insomnia which can lead many other problems like hypertension and dispersion. So, we can say that the WI-FI pollution is overall health risk. Therefore it needs to investigate the new innovation in this field which is present in the form of LI-FI [7] various wireless technology have different speed connection so that LI-FI have highest speed as shown in Table 1.

Table 1: Comparison of wireless technology

Technology	Speed
LI-FI	1Gbps
WI-FI	150 Mbps
Bluetooth	3 Mbps
IrDA	4Mbps
ZigBee	250 Kbps

II. LITRATURE REVIEW

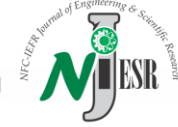
A. Existing wireless technology

The users of WI-FI have large in number up to millions to billions. No one can deny the importance and role of WI-FI at last century. But still we need to investigate the new technology

that just like WI-FI or may be opposite to WI-FI. Opportunities are numerous in every field but need to find the clue. Each and every organization and institutions used WI-FI communication medium for communication. The need of wireless technology increased in every field. Every network or some MAN have wireless router for communication. The reliable internet services in this world have major contribution of WI-FI. But instead of this why need to feel investigate the recent technology. The main reason is that all the wireless technology emits some harmful radiations. Due to take these radiations for very long time than the special disease occurs called insomnia. The insomnia basically creates the depression and hypertension. Usually practice is that all wireless devices used at night while using the device it's so near to us. Radiations and its strong effect directly or indirectly affect our brain nerves that change the sleep patterns. Anyhow, there are four fundamental reasons also:

- 1- Efficiency of a system
- 2- Availability of technology
- 3- Capacity of network
- 4- Security of technology

Dealing with all lights phenomena's no need to take an extra security problem. The availability and security then come to know the availability of light is everywhere. So, availability of light means available of LI-FI. LI-FI [8] operated on light in the form of LEDs and organic LEDs (OLED's) that uses visible light communication (VLC). If the LI-FI used inside the room than no one can know about either somebody used LI-FI or not. The other two factors like efficiency of LI-FI are 10,000 times greater than WI-FI in term of data density. Recently achieved the 224Gbps [9] . Haas says: LI-FI will be more secure due to its efficiency because WI-FI signal have ability to pass through walls due to radio signals so it may be demanding from neighbor to share your connection but on LI-FI hide signals by kept indoor for some curtains' etc. Haas also says that the LED bulbs could be dipped and dimmed on those points that they appear off, so it also still transmits data. Now at that time Hass's company started made LI-FI routers for a limited client and the name of this company is Pure LI-FI. May be for next several years they have large market areas and LI-FI becomes an integral part of industry and business educational institute. By this idea this concept easily implemented in any home appliances likes electronic kettle, iron and microwave; Internets of things (IOT) due to LI-FI have much faster than now. He also watches to control the remote locations of objects like roof top solar panels and hilltop transmitter. As LED Street lights, even be used on some roads and motorways that way the LED not only used for roads but also operated hotspot of LI-FI. For this the ratio of collision of V2V is minimal. So, the availability of LI-FI not only necessary for communication but also keep us to



secure of any haphazard situation. Now the question is appeared in our mind how this great technology is working? There is no any complex mechanism in which the overall LI-FI medium is depend but the simply use of VLC.

For this need a photo detector to receive light signals. WI-FI and LI-FI both transmissions of data electromagnetically but LI-FI use VLC while WI-FI uses radio signals. The bandwidth spectrum of radio waves is 0.4×10^{10} while VLC have such domain 4×10^{14} --- 7.9×10^{14} range value. As above spectrum easily predicts that LI-FI VLC is $\sim 10,000$ times greater than radio signals. All the LEDs [10] and photo detector equipment are easily operated on that spectrum. First, the photo detector received the light signals and then converted it into binary streaming context. All the light bulbs like LEDs are semiconductor material that conductivity lays between conductors and insulators. As semiconductor behavior, the LED bulb can be dimmed and dipped, down and up at very extremely high speed, even than with being too visible for human eye. As an example the data is fed into light bulbs LED [11] than send to photo detector. At this point the signal processing chambers convert it into binary stream able context. After it lamp driver attach with LED lamp that further processing the data electronically with some logic gates. If insert the amplification unit from received app data than can be perform the function of dipped/dimmed of LEDs. At the receiver side, the dongle will play role for device to maintain the signals. Figure 1 presents the diagrammatically working of LI-FI:

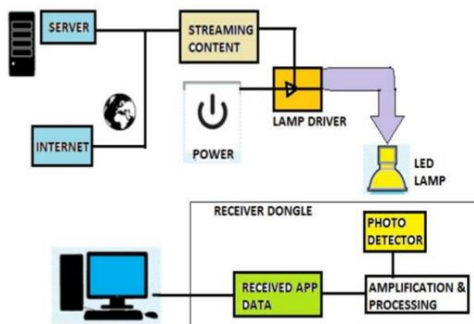
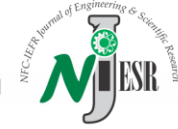


Figure 1: working mechanism of LI-FI [12]

WAVE another name of 802.11p is a reliable protocol in term of security and availability. Basically, 802.11p is a modified [13] form of 802.11a but it is very interesting both are based on orthogonal frequency division multiplexing (OFDM) [14]. WI-FI follow the IEEE standard 802.11b/g/n/ac/ad are also being accepted standards of WI-FI, while on other hand the WI-MAX that follow the IEEE standard that is 802.16 a/c/d/e.../m, but it uses propagation medium is microwave as compared to WI-FI that used radio waves. WI-FI uses the frequency of 2.4/3/5/60 GHz but 802.11p use 10 MHz

frequency [15] bandwidth that actually becomes the half of WI-FI frequency. To do this 802.11p have robust signal so that its tolerance increase for more than one propagation [14]. Here is the architecture of LI-FI in which show its main components and it's working in front of different phenomenon. The main and functional component that is basically oxygen of LI-FI is Dongle. The dongle is collection of distinct functions like amplification and processing the data stream through photo detector. So another famous name of LI-FI [16] is dongle based internet. 802.11b for WLANs is the famous name of wireless Ethernet. The beginning of WLANs in 1980s by the collection PCs and other distributed computer to share peripherals resources such as some access servers and pointers etc. The most popular local area network is Ethernet. WLANs and the family 802.11x support a high capacity LAN. From the 100meter of the base station WLANs support up to 11mbps data rate with 2.4 GHz band [17]. However, every base station can support many connections over a range of 100 meter above. WI-FI and LI-FI both are wireless, former is working with wireless router and latter is working with light. All the WI-FI enabled handheld devices, the issue of power consumption increases. By transmission and reception of data through web browsing the power consumed rapidly[18]. VoIP application can also be implemented over WI-FI [19] infrastructure mode. VoIP enabled phones to have similar with cell phone with infrastructure and use radio frequency. There are many chief advantages of these phones but talk time on these phones are very low. If our device VoIP application and connected some WI-FI routers than the performance of radiations become stronger and have some health issues. With connected WI-FI the talk time is reduced to only 1.5 hour maximum but when the WI-FI capability is turned off then it became 4 hour talk time minimum [18]. On the other hand, the LI-FI enabler devices have no issue of battery and power consumption etc. The major difference between LI-FI and WI-FI is direction of transmission. LI-FI, due to light medium strictly follows the (LOS) line of sight transmission while WI-FI follows non-line of sight (NLOS) transmission. The radius of LED [20] depends upon its size and shape, because of, light travel in a straight way and monochromatic in nature therefore LI-FI follows the LOS transmission. Those devices that are near the light source have robust signals in term of data density. That's why the data rate density comparatively very high as compared to WI-FI [21][4].

Another reason is that WI-FI [22] signal travel in all direction and penetrates the every object that suffers its performance like radio waves. So WI-FI [23] follow NLOS transmission its signal travel in all direction therefore a lot of signals are wasted during WI-FI routing signals, that is the reason WI-FI has low density as well as performance as compared to LI-FI [4]. A statistical example between LI-FI [24] and WI-FI is present when experiments take the result of LOS identification under



the area of 1500m. The experiments results show the LI-FI follow the LOS detection rate about 90.41% with some false alarm rate 9.35%. The numeric values based on temporal feature but in overall the spectral feature the LOS detection rate is 93.09% with a false alarm rate of 7.29%[25, 26].

B. LI-FI usage

According to the recent survey about the usage of LI-FI in different regions of the world the projected value that can be obtained from 2013-2018 is 84% [27]approximately throughout the world. This result is very amazing as compared to WI-FI; LI-FI hope so will be proved they as a best replacement of all existing wireless technology. The graphical representation show that the estimated value of LI-FI usage in different regions of the world. The Europe has leading state that use LI-FI product. For this current value easily determined if the usage of this technology become increase at a constant ratio then after some years the total estimated value reach the peak level. Its value is about 84% usage of LI-FI as compared to any other wireless device. [28]. Figure 2 presents the usage comparison of LI-FI in various parts of world.

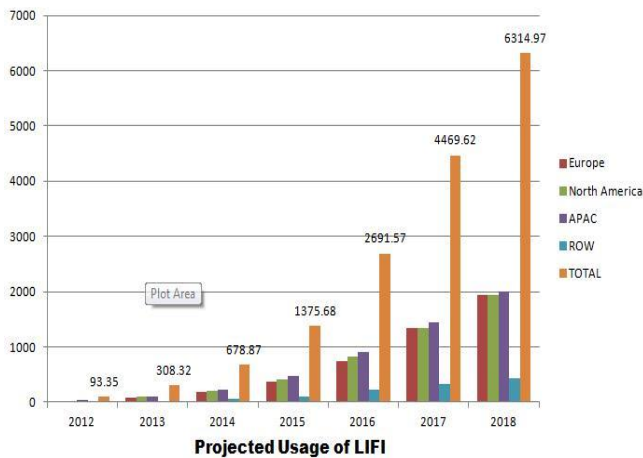


Figure 2: graphical representation of LI-FI usage

III. COMPARISON

Wireless communication based on various technologies that need distinctive characteristics, standards and techniques to provides the smooth services in the environment. Here is the block diagram of wireless communication standards i-e WI-FI and LI-FI. To show the clear difference take some common parameters between these two standards like cost, topology and medium etc. All the main categories are further divided in sub-category to make the difference of technology is very clear but several parameters are shared because both of these technologies used. But it is very clear from the below diagram the role of VLC cannot be ignored in case of OWC that have long range and FSO that have short range. Figure 3 presents

various standards and different parameters that are needed by WI-FI and LI-FI.

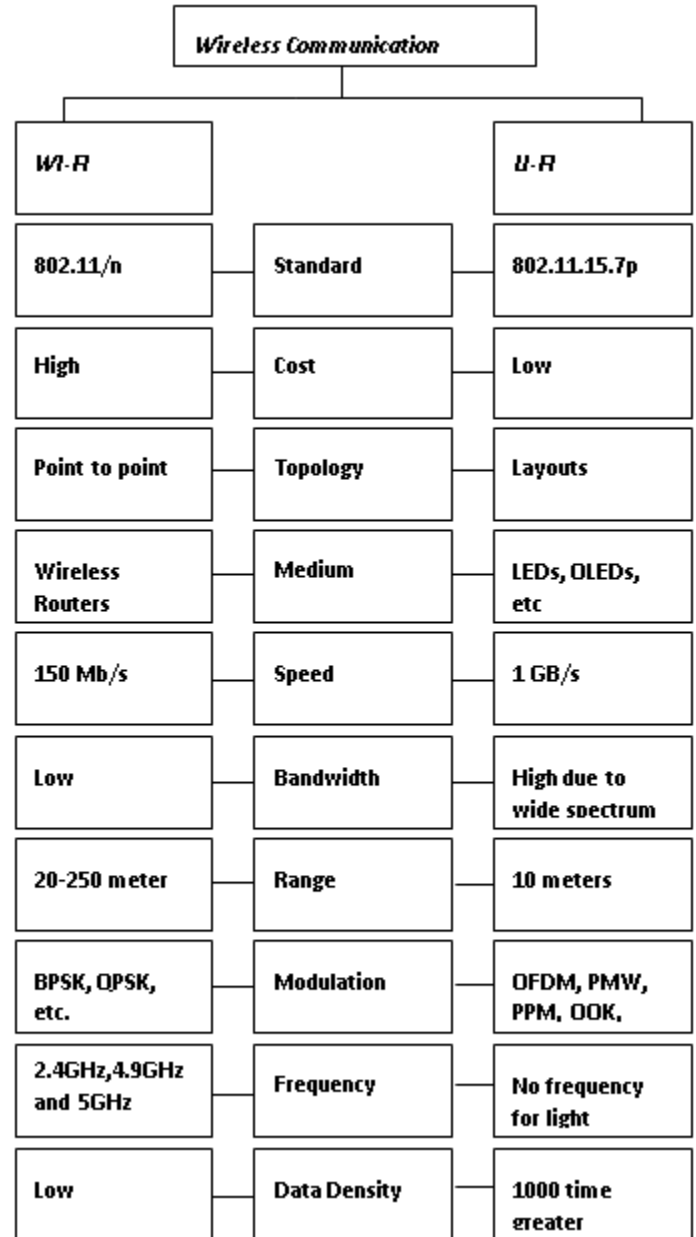


Figure 3: Comparison based on different parameters

A. Comparison between technologies

Here is the detail comparison between two technologies i-e; VLC and IR respectively. The LI-FI used VLC and WI-FI used IR as shown in figure 4.

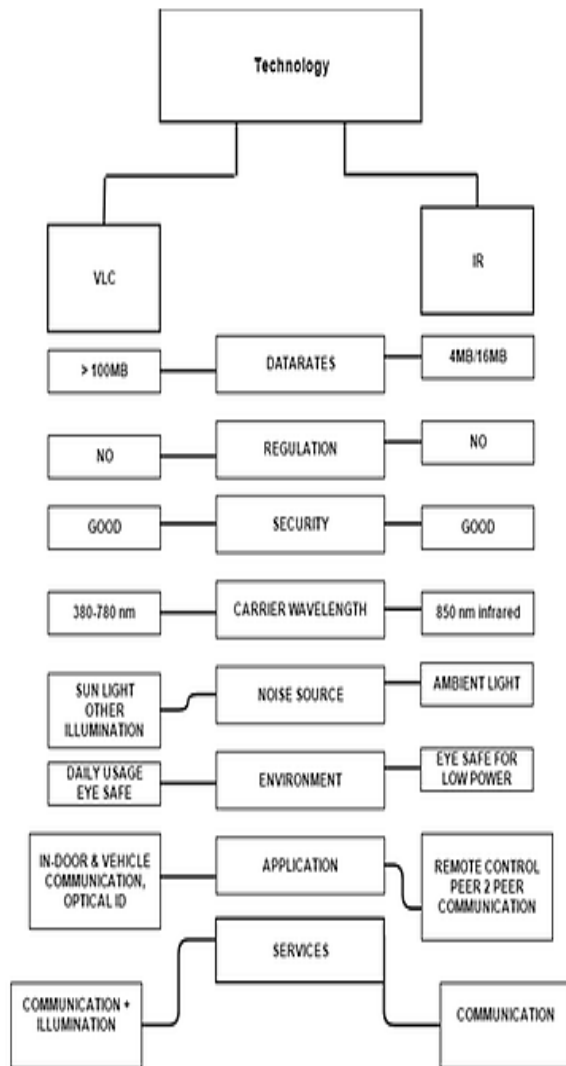


Figure 4: Comparison between LI-FI's and WI-FI's technology

B. Challenges of LI-FI

LI-FI has a lot of advantages over WI-FI but LI-FI technology have also some problems that are directly and indirectly affect the performance of LI-FI[29]:

- Interference of external light like sun, normal light bulbs, opaque materials etc.
- Biggest challenge is how receiving device will support uploading signals continuously.
- Can't pass through objects

- Basically LI-FI based on VLC, it can't penetrate through walls bricks as radio waves and is easily blocked by objects
- VLC require line of sight because it is unidirectional mean P2P light communication. While LI-FI range doesn't require line of sight between the receiver and transmitter.
- Major challenge is LI-FI range mean escape out the circle and radius of light simply means leave the communication connection.

C. Application of LI-FI

Due to light supported technology there are the various application of LI-FI that not implemented yet. All the application of LI-FI can be extended where WI-FI technology lacks its presence like power plants, medical technology and nuclear sector. Since LI-FI uses only light for communication so it safely be used where WI-FI banned because its prone to interfere with the radio waves.

Here is the some application[30] of LI-FI that can be implemented in near future as following:

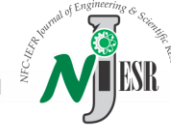
Underwater communication: The radio waves could not properly propagate under water so WI-FI might be failed for underwater [31] communication but LI-FI is very suitable for any type of underwater communication.[32, 33].

Education system: Today education institute play vital role in any type of research. If the entire educational institute are embedded with LI-FI than every class room and conference room are a workshop of internet. Everyone knows the major problem in educational institute is fast and high speed of internet so by using LI-FI this problem easily removed.

Availability of internet anywhere: Since light is present everywhere so internet is present everywhere. By intelligently used this technology every light bulb, street lamps, lights of vehicles used to access availability of internet everywhere like footpaths, roads, shopping malls, each and every where light source is available.[34].

Traffic management system: Today we spent half of our life in traffic problems. The main issue is lack of traffic management. If anyone violate the traffic signals than it's seemed to be impossible to catch it. If the use the traffic system in the hands of LI-FI[35] than everything not only automatic but also easy to trace then location of volunteer that violate the traffic signals by use of update traffic information with the help of light.[36]

In healthcare sector: Although radio signals are very robust in nature but overall, it's dangerous for human body because it penetrates human body so it's not safe to be used in any health sector like hospitals and some various health care sector. On the



other hand LI-FI [37] can be implemented in a well-disciplined manner in any health care sector.[12]

In sensitive area: WI-FI is not a good option in any secret area like nuclear power plants and radiator sector etc. because WI-FI having the issue of security and with propagation of signal in all around so can easily hacked. Any sensitive areas LI-FI can provide safer communication that never be hacked even nobody know about the usage of any internet device in this particular sector [38].LI-FI can also be used in chemical plants, petroleum industry, mining industry etc. where other any type of transmission medium is unsafe.

In disaster management: In any tragedy situation like hurricanes, earthquake, LI-FI bulbs are powerful means of communication. It provides cheap access to every corner for quick response. So, this communication technology is helpful for emergency areas where WI-FI signals not easily be reached and deployment of such network is difficult.

Replacement for other technologies: LI-FI does not used radio waves so it is easily being used in those areas where other technologies like Bluetooth, WI-FI, infrared etc are banned.

IV. CONCLUSION

Both technologies have some merits and demerits to some extent. Towards green communication the role of LI-FI discovery has been considered. The role of LI-FI cannot be ignored because recently its application in every field of life has numerous advantages. Currently LI-FI is not a longer concept and an idea but a future technology. Due to still in its infant age it has vast potential. A lot of research can be conducted still now to improve its working strategy and explore VLC system. So, the role of VLC in this technology has some fruitful effort. But still have a lot of possibilities and can be explored further. In some real time, implementation try to make this technology in its practical field then every light bulb is now be a part of LI-FI system as represent like WI-FI hotspot to transmit any type of wireless data. This survey paper provides a lot of detail about comparative study between these two technologies. In this paper, it has been observed that LI-FI is most superior with respect to data transmission over WI-FI technology and it provide a pathway towards IOT devices for valuable communication. Up to our best knowledge no latest survey would be conducted on the behalf of specific parameters Simulation, validation and testing modulation technique can be a part of future work. With the LI-FI technology we will move towards the greener, safer, healthier, and cleaner future of wireless technology.

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